

SPRING 2026
FINCB9325 Financial Econometrics – Time Series
Homework 3
Geert Bekaert

Question 1:

The distribution of stock returns has fat tails (see lecture 1). As a result, we need models that deliver fat-tailed distributions (e.g., to manage risks, to price options on stocks). One way of doing this is to introduce jumps or crashes in the model. This problem describes the building blocks of how to do that.

Consider a Bernoulli-distributed variable:

$$B_t = \begin{cases} 0 & \text{with probability } 1 - p \\ 1 & \text{with probability } p \end{cases}$$

and two independent Standard Normal variables ε_t and δ_t . Define the jump as

$$J_t = B_t \times (\mu_J + \sigma_J \delta_t)$$

and let log returns follow the process:

$$r_t = \mu + \sigma \varepsilon_t + J_t.$$

Assume B_t is independent of ε_t and δ_t .

1. Derive the population mean, variance, skewness, and excess kurtosis of this distribution.
2. Explain carefully why this Bernoulli-normal mixture is potentially a better model of returns than the log-normal model.
3. Suppose that log returns r_t are simply i.i.d. distributed $\mathcal{N}(0.0004, 0.01^2)$. Plot a simulated series of 12,500 daily observations (50 years). Does it look like the data? What is missing? Next, suppose that log returns r_t are given by the above jump model with the following 5 parameters:

$$(\mu, \sigma, p, \mu_J, \sigma_J) = (0.0004, 0.01, 0.01, -0.03, 0.04).$$

Using these parameter estimates, what are the unconditional mean, standard deviation, skewness and kurtosis of the simulated log stock returns? Again, plot a simulated series of 2500 observations ($T = 10$ years). Does it look like the data? What is missing?

4. Here, we will assess the properties of the asymptotic White standard errors (no autocorrelation, but heteroskedasticity and non-normality robust standard errors). Simulate N samples

from the return process with jumps in subquestion 3 with $T = 1$ (250 observations), where T indicates years (and we assume 250 trading days per year). Within each of the N samples, calculate the sample mean and construct the “t-stat” of the sample mean, under the null that the population mean equals zero, using the central limit theorem from the notes and the (correct in this case) assumption that there is no autocorrelation in returns to construct the standard error. Give the mean, standard deviation, skewness and kurtosis of the N “t-stats” and sample means you have constructed. Do the sample means and “t-stats” look normally distributed? Compare the average standard error across the N samples to the standard deviation of the N sample mean estimates. Compare and discuss.

Repeat this analysis for $T = 0.10, 0.50, 2$, and 4 years’ worth of daily data and note any patterns of interest.

Question 2:

Download a set of monthly data from Canvas. The variables are:

- `div_yld`: annual dividend yield
- `log_excess_ret`: log excess returns

1. Run a simple one-step-ahead predictability test through OLS:

$$12 \times r_{t+1} = \alpha + \beta dy_t + e_{t+1}$$

Use “robust” standard errors (controlling for heteroskedasticity, so-called “White” standard errors) and test the null that $\beta = 0$. You can express both in actual percent, so e.g. 0.03 for the dividend yield, and 0.15 for the returns. By multiplying the return by 12, the regression coefficient indicates how much percentage points a one percent (absolute) change in the dividend yield (e.g., from 3% to 4%) changes the annualized equity premium.

2. Is the OLS estimator in (1) consistent? Explain/prove.
3. Create $r_{t+k,k} = r_{t+k} + r_{t+k-1} + \dots + r_{t+1}$. Set $k = 12$. Run the same regression but with $r_{t+12,12}$ as the dependent variable, with monthly data. These data have an overlapping structure. What process does the residual follow? How would you adjust for that in your standard errors? Again, run the test that β is zero. Also, report the R^2 .